Fiscal Year:	FY 2017	**Task Last Updated:**	FY 02/27/2017
PI Name:	Snyder, Michael Ph.D.		
Project Title:	HERO Twin Astronaut Study Consortium (TASC) Project: Longitudinal integrated multi-omics analysis of the biomolecular effects of space travel		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94305-2200	**Congressional District:**	18
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-TWINS
Start Date:	04/16/2014	**End Date:**	04/15/2018
No. of Post Docs:	3	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Extended to 4/15/2018 per PI and NSSC information (Ed., 4/21/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Winkelmann, Juliane M.D. (Stanford University)		
Grant/Contract No.:	NNX14AH52G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Key Objectives: The rigors of space travel present conditions for which our species have not adapted, and thus it would be a significant public health benefit to characterize physiologic and cellular responses to the unique environmental conditions associated with space travel including solar radiation, prolonged weightlessness, G-transitions, and circadian disturbances. Our goal is to perform a longitudinal integrative analysis of all omics data generated during the Human Exploration Research Opportunities (HERO) mission to provide an unprecedented portrait of the molecular and physiological changes that accompany space travel. Methods/Techniques: The Snyder laboratory at Stanford has recently pioneered longitudinal integrative personalized omics profiling (iPOP), which combines genome-wide sequencing with the regular sampling of a patient's transcriptome, proteome, metabolome, and cytokine profiles in order to detect multi-analyte signatures that correlate with disease onset and progression. Our approach will be to leverage our unique expertise in integrated omics studies to combine the data generated at each of the HERO sites in an integrated analysis framework to detect longitudinal multi-analyte signatures that originate specifically during or after the space mission. Using data-driven network inference, and knowledge-based pathway databases, we will build quantitative models of molecular network activity, observe temporal dynamics of network activity, and identify network modules associated with space travel. We will determine flight-specific biomolecules and pathways and detect direct, inverse, and phase-shifted correlations between flight-specific clinical and molecular signatures and significant events recorded during the space mission such as G-transitions during takeoff and landing, recorded solar events, etc. We are also collecting untargeted metabolomics and proteomics profiling data that will be a very useful complement to the other data types being collected, and will provide biological insights on physiology and metabolism associated with space flight. We hypothesize that through integrative omics analysis we will discover biological signatures associated with G-transitions, prolonged weightlessness, DNA damage response to solar radiation, and other environmental changes that occur during the space mission. The Snyder lab is also responsible for coordinating data transfer and storage by providing a secure environment for data storage and setting up secure data transfer methods.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed work is highly innovative and will provide the first large-scale omics data ever collected on a human during space travel and will present an unprecedented longitudinal analysis of monozygous twins placed in vastly different environments. This comprehensive biological analysis will allow us to better understand the health risks associated with space travel and develop strategies to mitigate these risks.
Task Progress:	We have already generated untargeted metabolomics data on plasma samples, and are in the process of generating untargeted proteomics data. Preliminary analyses of the metabolomics data have yielded some interesting findings that were presented at the 2017 NASA Human Research Program (HRP) Investigators' Workshop. We have developed a two-step feature selection and prediction method that can handle multi-omics data in an integrated manner, and are currently testing the robustness of the method. We are also investigating alternative data-driven network based methods, and will apply them to the Twins Study data once data generation is complete. We have also set up a secure, dbGap-compliant server with two-factor authentication for storage of the data, as well as a secure encrypted data transfer system, which has already been used by some Twins Study groups to transfer data.
Bibliography Type:	Description: (Last Updated: 08/12/2021)
Abstracts for Journals and Proceedings	Mishra T, Piening B, Contrepois K, Ahadi S, Salins D, Snyder MP. "Longitudinal integrated multi-omics analysis of the biomolecular effects of space travel." 2017 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 23-26, 2017. 2017 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 23-26, 2017. , Jan-2017